CS-302: Artificial Intelligence

LECTURE NO 9:

Organization of Lecture

1. Single-State Problem Formulation V/S Multiple-State Problem Formulation.
2. Default Reasoning.
3. Monotonic Reasoning V/S Non-Monotonic Reasoning.
4. Production System in A.I.

1. Single-State Problem Formulation V/S Multiple- State Problem Formulation (1/2)

1) Single-State Probelm formulation V/S Multiple-state Probelm formulation

Single-State Probelm formulation	Multiple-State Probelm formulation
↳ Exact Prediction is possible	Semi-exact prediction is possible
State → is exactly known after any action sequence.	State → not exactly known, but limited to a set of possible states.
Accessibility → Information achieved through sensors.	Accessibility → not all essential information can be achieved through sensors. [Reasoning]
Consequences → of actions are known to agent.	Consequences → Not always known to the agents [Randomness]

1. Single-State Problem Formulation V/S Multiple-State Problem Formulation

Goal → unique for each known initial state.	Goal → No fixed action sequence that leads to Goal
↳ [Simple but restricted] → e.g: Vaccum World	↳ Less restricted but more Complex. → e.g: Vaccum world but no sensors.
↳ It can't deal with incomplete accessibility [Limitation] ↳ Incomplete knowledge about the consequences can alter the world [Limitation]	↳ Can't deal with changes in the world during execution [Limitation]

2. Default Reasoning (1/2)

2) Default Reasoning

- It is very common form of non-monotonic reasoning
- Conclusions are drawn based on what is most likely to be true.
- Approaches to Default Reasoning → (Non-Monotonic Logic and Default Logic)

2. Default Reasoning (2/2)

Non-Monotonic Logic

↳ Truth of Proposition may be changed when new information are added and logic may be built to allow the statement to be retracted.

↳ Modal operator (M) → Consistent with everything we know

Examples:

→ Fact 'F' is true, we can attempt to prove '~F'

↳ Fail ∴ F is True

→ $\forall x$: plays-instrument(x) ∧ M manages(x) → jazz-musician(x)

For all x

If x plays an instrument

and

x manages it consistently

then

Conclusion

x is a jazz musician

Default Logic

↳ Initiates a new Inference Rule

$$\frac{A : B}{C}$$

A: Pre requisite

B: justification

C: Consequent / conclusion

[If A and if it is consistent with rest of what is known to assume that B, then conclude that C.]

Examples

$$\frac{BIRD(X) : FLY(X)}{FLY(X)}$$

$$\frac{BIRD(Tweety) : FLIES(Tweety)}{FLIES(Tweety)}$$

if tweety is a bird and it is consistent that tweety can fly then tweety can fly.

3. Monotonic Reasoning V/S Non-Monotonic Reasoning

3) Monotonic Reasoning V/s Non-Monotonic Reasoning

Monotonic Reasoning: Once the conclusion is taken, then it will remain same even if we add some other $info^n$ to existing $info^n$ in our knowledge base? As decisions are not affected by new facts, not suitable for real-time system

($info^n$ 1) ⟶ Decision
+
⋮

E.g.:
↳ Earth revolves around SUN] earth is not round
New $info^n$

↳ All old proofs are valid.] Adv

↳ Can't represent real-world scenarios] Disadv

↳ New knowledge from real-world can't be added] Disadv.

Non-Monotonic Reasonings: In this some conclusion may be invalidated if we add some more $info^n$ to our knowledge base.

$info^n$ 1 ⟶ d1, d2 } Decisions can be changed by new facts.
+
$info^n$ 2

↳ Helpful in Real-world scenarios.] Adv.

E.g.: Birds can fly
Penguins can't fly
Alex is a bird] Alex can fly
+
Alex is a Penguin] Alex can't fly

Disadv.
↳ Can't be used for theorem proving.

4. Production System in A.I (1/3)

4) Production System in A.I

↳ Helps in structuring A.I Programs in a way that facilitates describing and Performing the search process.

↳ Production System consists of '4' parts:

(i) Set of rules (if-then etc)

(ii) Knowledge Base (Data base)

(iii) Control Strategy (It specifies the order in which rules need to be compared to the database so that the conflict is resolved in a minimum time).

(iv) Rule Applier (Rules are applied upon control strategy).

4. Production System in A.I (2/3)

Steps to solve the Probelm

i) First reduced the probelm so that it can be shown in a imprecise statement.

ii) Problem can be solved by searching a path through space. [Start State $\longrightarrow$ Goal state].

iii) Solving process can be modelled through a production system.

Advantages

i) Structuring A.I Problems] Ex cellent Tool.

ii) Highly Modular] rules can be added, removed or changed.

iii) Rules are expressed in natural form] Rule is easy to understand.

4. Production System in A.I (3/3)

Characteristics:

✔ i) Monotonic Production System: Application of a rule never prevents later application of another rule] Rules are independent

ii) Non-Monotonic Production System: which is not true

✔ iii) Partially Commutative Production System:

r_1 = rule 1
r_2 = rule 2

Allowable

(r_1, r_2)

$x \to y$

permutation $\to (x \to y)$ (Any combination of rules r_1 & r_2)

iv) Commutative Production System:

↳ It is both monotonic and partially commutative production system.

Assignment:

Analyze and explain working of

1. Vaccum World using single state
2. Missionaries and cannibals problem
3. N- queen problem

References: Textbooks

1. *"Artificial Intelligence", by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
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3. *"Computational Intelligence: A logical Approach", by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
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5. *"AI: A new Synthesis", by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *"Artificial Intelligence: Theory and Practice" by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*